

EXPERIMENT NO. 10 / 11

Software Configuration Management (SCM) & Git Version Control

Project: Hospital Management System (HMS)

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1. Aim

To study Software Configuration Management (SCM) concepts and develop different versions of the Hospital Management System (HMS) specifications and code using Git, tracking baselines, branches, and change control procedures.

2. Objectives

- To understand the core SCM processes: Identification, Version Control, Change Control, Configuration Auditing, and Reporting.
- To identify Configuration Items (CIs) in the HMS project (SRS, DFD/UML models, database scripts, source code).
- To manage version branching and merge conflict resolution in Git.
- To simulate a formal Change Request (CR) workflow reviewed by a Change Control Board (CCB).

3. HMS Configuration Items (CIs) & Baseline

CI Identifier	Configuration Item Name	Baseline Version	Responsible Role
CI-01	IEEE 830 SRS Document	v1.0 (Frozen)	Product Owner / Systems Analyst
CI-02	UML & DFD Architectural Models	v1.0 (Approved)	Software Architect
CI-03	MySQL Relational Schema (DDL scripts)	v1.1 (Staging)	Database Administrator
CI-04	Hospital Billing Core Module (calculateBill)	v1.2 (Tested)	Lead Backend Developer
CI-05	Payment Gateway REST API Integration	v1.3 (Integrated)	Integration Engineer

4. Git Branching Strategy & Workflow Commands

The HMS team utilizes a Git-flow branching strategy consisting of:

- main: Production-ready releases tagged with semantic versions (e.g., v1.0.0).
- develop: Integration branch where completed feature branches are merged.
- feature/patient-registration: Branch for patient registration UI and token generator.
- feature/billing-gateway: Branch implementing calculateBill() and UPI integration.
- hotfix/billing-rounding-error: Emergency bug-fix branch created from main.

Git Command Trace:

```
# 1. Initialize repository and create initial commit
git init HMS_Repository
cd HMS_Repository
git add .
git commit -m "feat: Initial commit of HMS SRS and system specifications"

# 2. Establish development branch
git branch develop
git checkout develop

# 3. Create and work on Billing feature
git checkout -b feature/billing-module
# [Edit HospitalBillingTest.java]
git add HospitalBillingTest.java
git commit -m "feat(billing): implement calculateBill() with insurance waiver logic"

# 4. Merge feature into develop
git checkout develop
git merge --no-ff feature/billing-module -m "merge: integrate billing module into develop"

# 5. Release tagging on main
git checkout main
git merge develop
git tag -a v1.0 -m "Release v1.0: Baseline approved by CCB"
git push origin main --tags
```

5. Conclusion

Using Git for Software Configuration Management ensures full traceability, version reproducibility, and secure collaboration across all HMS configuration items.